

Martin Nguyen

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EDUCATION

University of Toronto

Mississauga, ON

Bachelor of Science in GeoSpatial Data Science & Statistics

Sept 2024 - Present

- Relevant Courses: Machine Learning, Methods of Data Analysis, Spatial Data Science, Computational Statistics, Probability & Statistics II, Software Design, GIScience

EXPERIENCE

Deep Learning Perception Analyst

Feb 2026 - Present

University of Toronto Formula Racing Team

Toronto, ON

- Conducted a MobileNetV2 cone-classification study across 5 input resolutions (32–128 pixels), tracking validation accuracy, macro-F1, and loss in Weights & Biases.
- Adapted Python training code and built a SLURM launcher to run 5 GPU experiments of 20+ epochs each on the Nibi cluster.
- Annotated 60+ track images in Roboflow and reviewed 50+ images, correcting bounding boxes and cone classifications to support perception model development.

Signal Operator

June 2025 – Present

Canadian Armed Forces

Toronto, ON

- Supported field communications across 20+ tactical radio and data devices by diagnosing and resolving frequency, configuration, and connectivity issues.
- Established operational communications setups by integrating radios, antennas, generators, and cryptographic equipment across 4 equipment categories.
- Maintained accountability for \$5000+ worth in assigned communications equipment, following security and handling procedures during field operations.

Mobile Developer Intern

July 2025 - Oct 2025

Surveil.One

Remote

- Developed 20+ Flutter screens for an AI-powered monitoring dashboard, translating product requirements into functional mobile workflows.
- Integrated GraphQL APIs to retrieve and display backend data within the Flutter application, connecting dashboard interfaces to backend services.

PROJECTS

NBA Space Explorer | *JavaScript, React, Vite, SVG*

- Built an interactive basketball shot chart using React and SVG to visualize shot locations and distinguish made and missed attempts.
- Implemented player and team filters to explore shot distributions and compare scoring patterns across court locations.
- Developed a shot data loading and normalization pipeline and generated a 500-shot sample dataset for demonstrating the application.

NorthStar | *TypeScript, React, Next.js, Mapbox GL JS, GeoJSON*

- Collaborated on a mapping application that uses Toronto crime data to help users compare routes based on travel time and estimated risk.
- Contributed to route scoring supporting 3 preferences: fastest, lowest estimated risk, and a balance of time and risk.
- Helped integrate crime heatmaps and Google Street View into an interactive Mapbox map, allowing users to inspect incidents and surrounding streets.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, JavaScript

GIS & Visualization: ArcGIS, Mapbox GL JS, GeoJSON

Data & Tools: PowerBI, Tableau, Excel, tidyverse, RStudio, Roboflow

Frameworks & Development: React, Next.js, Flutter, GraphQL, Git, GitHub